MIL-U-36000B
2 April 1976
SUPERSEDING
MIL-F-36000A
11 March 1971

MILITARY SPECIFICATION

UNDECYLENIC ACID AND ZINC UNDECYLENATE POWDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Undecylenic Acid and Zinc Undecylenate Powder (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Box, Setup
PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (see 6.3)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated,

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the issue in effect date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a uniform, homogeneous, grit-free powder mixture containing:

Undecylenic Acid	2 percent
Zinc Undecylenate	20 percent
Talc	78 percent

3.1.1 Assay for total undecylenic acid content. Total undecylenic acid content shall be not less than 17 percent and not more than 21 percent by weight when assayed as specified in 4.5.1.

3.1.2 Assay for free undecylenic acid content. The free undecylenic acid content shall be not less than 1.5 percent and not more than 2.5 percent when assayed as specified in 4.5.2.

3.1.3 Iodine number. The iodine number shall be not less than 22.9 and not more than 28.5 when determined as specified in 4.5.3.

3.1.4 Sieve test. Not less than 99 percent of the foot powder shall pass completely through a No. 100 U.S. Standard sieve, or equivalent, and not less than 95 percent of the foot powder shall pass through a No. 200 U. S. Standard sieve, or equivalent when tested using the U.S.P. method for powder fineness.

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3.2 Ingredients. All ingredients entering into the finished powder shall be of U.S.P. quality. All other ingredients entering into the finished powder shall be of U.S.P. or N.F. quality or, if not included in either of these compendia, the ingredients shall be suitable for use in this preparation.

3.3 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the powder, to the date of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of powder produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

Table I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container.

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4.3.1 For end item testing. Noncompliance with any of the following is cause for rejection.

Table II. End item testing

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample	
Assay for total undecylenic acid content	3.1.1	S-2	5	Composite
Assay for undecylenic acid content	3.1.2	S-2	5	
Iodine number	3.1.3	S-2	5	
Odor test	3.1.4	See U.S.P.	-	

Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished powder shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.5.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects

Categories	Defects 1/
Major	
101	Powder not free from foreign matter.
102	Powder not grit-free.
103	Powder not homogeneous.
104	Sifter top or sifter holes not functioning properly.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Tests.

4.5.1 Assay for total undecylenic acid content. Into a 250-ml beaker, accurately weigh approximately 4 grams of the sample. Add 100 ml of purified water and 15 ml of stronger ammonium hydroxide solution, mix thoroughly, and allow the powder to subside. Decant the liquid through a filter into a 500-ml separatory funnel. Add 100 ml of purified water and 10 ml of stronger ammonium

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hydroxide solution to the residue in the beaker, mix and filter as specified above. Wash the residue with approximately 75 ml of purified water and filter into the separatory funnel. Discard the residue. Add 35 ml of concentrated hydrochloric acid to the filtrate in the separatory funnel. Extract with one 100-ml portion and two 75-ml portions of petroleum ether (boiling point 30° - 60° C.), combine the ether layers and wash with purified water until free of mineral acid. Transfer the petroleum ether to a 500-ml Erlenmeyer flask and evaporate on a steam bath to approximately 5 ml. Add 100 ml of alcohol neutralized to phenolphthalein and titrate with 0.1N sodium hydroxide using phenolphthalein as indicator. Each ml of 0.1N sodium hydroxide is equivalent to 0.01843 grams of undecylenic acid.

4.5.2 Assay for free undecylenic acid. Into a 250-ml beaker, accurately weigh approximately 20 grams of the sample. Add 100 ml of petroleum ether (boiling point 30° - 60° C.), mix thoroughly, and allow the powder to subside. Decant the liquid through a filter into a 500-ml Erlenmeyer flask. Wash the residue with three 50-ml portions of petroleum ether, mix, and filter as specified above. Evaporate the petroleum ether filtrate on a steam bath to approximately 5 ml. Add 100 ml of alcohol neutralized to phenolphthalein and titrate with 0.1N sodium hydroxide using phenolphthalein as indicator. Each ml of 0.1N sodium hydroxide is equivalent to 0.01843 grams of undecylenic acid.

4.5.3 Iodine number. Into a tared weighing cup, accurately weigh approximately 1.1 grams of sample and transfer it quantitatively to a 500-ml iodine flask. To both this and a similar flask, add 20 ml of carbon tetrachloride. Mark the flask without the sample as the blank. Add volumetrically 25 ml of Hanus iodine solution to each flask, stopper, and store in the dark for 30 minutes. Add 20 ml of 15 percent potassium iodide solution and 100 ml of purified water and back titrate the unused iodine with 0.1N sodium thiosulfate solution.

Calculation:

$$\frac{(\text{ml Blank} - \text{ml titration of sample}) \times (\text{N of } NA_2S_2O_3) \times 12.69}{\text{Sample Weight}} = \text{Iodine Number}$$

4.6 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units. For labeling, the acceptance number shall be zero.

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TABLE IV. Examination for preparation for delivery 1/

Examination	Defect
Containers	Not as specified
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging & Packing Material	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

1/ The examination is only applicable to the intermediate and exterior containers.

5. PACKAGING

5.1 Packaging requirements. Requirements for packaging shall be in accordance with all applicable paragraphs of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One can containing 45 grams (1.5 ounces), as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available sifter top can considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatability that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. A unit package shall not be supplied for each immediate container.

5.2.2.3 Intermediate package. Six immediate containers shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening.

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5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.3.1 Level A. Twenty-four intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636 class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.3.2 Level B. Twenty-four intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commerical practice.

(b) the date of manufacture prefixed by "MFD".

(c) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed.

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186 except that the date of manufacture shall be shown in lieu of the date packed.

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DOCUMENT IDENTIFIER AND TITLE

MIL-U-36000B UNDECYLENIC ACID AND ZINC UNDECYLENATE POWDER

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